

MAJOR ELECTIVES

SEMESTER V

Programme	B. Sc. Statistics				
Course Code	STA5EJ301				
Course Title	Statistical Quality Control				
Type of Course	Major Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary Objective	To make students aware of Various Quality or standards in Industrial Production, Detecting, Controlling and Maintaining Quality and Total Quality Management				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understanding notion of Quality of products	U	C	Instructor-created exams / Quiz
CO2	Assessing various meaning of Quality	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Explain causes of variation and Statistical Control	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Construction of Control Charts and OC	U	C	Instructor-created exams / Quiz

	curves			ated exams / Home Assignments
CO5	Distinguish Process and Product Control	Ap	P	One Minute Reflection Writing assignments
CO6	Assessing Process and Product Control	Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Control Charts theory		10	15
	1	General theory of Control Charts.		
	2	Setting Control Limits.		
	3	Importance of 3-sigma limits		
	4	Statistical basis of Control Limits		
	5	Need of two control charts for variables		
	6	Assessing Statistical Control using Charts		
	7	Control Charts for Variables and Attributes		
	Sections from References:			
II	Control Charts Construction		14	20
	8	Mean Chart Theory and Construction		
	9	Dispersion (Range, Standard Deviation Chart) Chart. Theory and Construction		
	10	Proportion defective Chart Theory and Construction		
	11	Number of Defective Chart Theory and Construction		
	12	Number of Defects Chart Theory and Construction.		
	Sections from References:			
III	Product Control		14	20

	13	Sampling Inspection Plans (Acceptance Sampling Plans)		
	14	Single Sampling Plan		
	15	Double Sampling Plan, ,		
	16	Sequential Sampling Plan		
	17	Incoming and Outgoing Quality		
	18	AQL, RQL, LTPD, AOQ, AOQL		
	19	Errors in Sampling Inspection Plans		
	20	Power function and OC function.		
	21	Producer' and Consumers Risk		
	Sections from References:			
IV	Characterising Sampling Plans		10	15
	22	Constructing OC Curve of Single Sampling Plan using Hyper Geometric distribution		
	23	Constructing OC Curve of Single Sampling Plan using Binomial distribution		
	24	Constructing OC Curve of Single Sampling Plan using Poisson distribution		
	25	Constructing OC Curve of Double Sampling Plan		
	26	ASN, ATI		
	Sections from References:			
V	Open Ended Module:		12	
	1	Preliminaries of Quality Control Definition of Quality. Need of total quality Management and its uses. Causes of Variation, assessing within and between sample variation using Statistical Measures. Concept of Statistical Quality Control, Process Control and Product Control		
	Sections from References:			
Books and References:				
1. Introduction to Statistical Quality Control, 8th Edition Douglas C Montgomery				
2. Statistical Quality Control M Mahajan Dhanpat Rai 2nd Edition				
3. Fundamentals of Applied Statistics S C Gupta and V K Kapoor Sultan Chand & Sons				